2.1.3 Effects of maternal malnutrition on the foetus

a) Congenital malformations in the foetus

Foetal wastage is high in low- income population, one can assume that dietary deficiencies do contribute to higher incidence of congenital malformations. The incidence is much higher if mothers have suffered from viral infection such as rubella, influenza etc.

b) Birth weight

The birth weight of infant is influenced by many factors such as maternal age, parity, height, altitude, ethnic origin and socioeconomic status. Since mother's belonging to low income groups are lighter and shorter, it may be argued that the low birth weight observed in this population is an effect of maternal size, which in turn is an outcome of poor nutritional status of the mother.

c) Infant mortality

The major components of infant mortality i.e. perinatal (28 weeks of gestation to 7 days postnatal) and early neonatal (7 days - 1 month after birth) mortality, are directly related to the health and nutritional status of the mother during pregnancy. Due to high incidence of low birth weight and prematurity in poor - communities perinatal and neonatal death rates are also higher and contribute to almost 60 % of infant deaths.

d) Development of brain and mental function

The peak period of human brain growth is in the last few weeks of intrauterine and first 6 months of extra uterine life. After this, the brain growth slows down, nutritional insults during these phase can be expected to affect brain development and lead to poor mental function.

e) Delayed consequences of foetal growth retardation

In recent years, it is increasingly recognised that effects of foetal malnutrition continues into the adult age, if the infant survives. It is observed that the incidence of chronic metabolic disease like diabetes, hypertension, cardiovascular insufficiencies and cerebrovascular stroke are more common in these infant when grown into adulthood than the normal birth weight adults.

2.1.4 Nutrition requirement during pregnancy

a) Calories

Energy requirement during pregnancy is increased for maintaining the growth of the foetus, placenta and maternal tissues and for the increased basal metabolic rate (BMR). The additional energy cost of pregnancy for a 55 kg woman has been estimated to be around 80,000 kilocalories. The energy needs are evenly distributed throughout the pregnancy. Energy requirements are also influenced by the pre pregnancy body weight, physical activity and age. WHO recommends an additional 150 Kcal/day in the 1st trimester and 350 Kcal/day in the last 2 trimesters.

b) Protein

The additional protein requirement during pregnancy is mainly due to accretion of protein by the foetus, the enlargement of uterus, mammary glands, placenta, formation of amniotic fluid and storage reserves for labour, delivery and lactation which is around 1000g for the entire pregnancy. For this additional daily requirement, allowance of a good quality protein like 10g milk or egg protein per day has been suggested. The protein required for a pregnant woman is 82.2g/ day. Women who are chronically undernourished and underweight, those with infection and infestation and adolescent pregnant women, may require extra protein in the form of milk or other animal proteins and calories for repletion of tissue proteins to enable them to with stand the stress of pregnancy and lactation.

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